Website Risk Analysis – Results Explanation

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1. Introduction

This project focuses on evaluating merchant websites to identify potential risk signals before onboarding. The goal is to simulate how fintech platforms perform an initial risk assessment using a rule-based system.

The solution combines on-site analysis (content, structure, and signals within the website) with basic external intelligence (domain-level and reputation signals). Instead of using complex machine learning models, the system is designed to be simple, explainable, and practical for real-world use cases.

2. How the System Works

The system evaluates each website in four stages:

- First, it extracts key elements such as visible content, hyperlinks, and contact information (emails and phone numbers).
- Next, rule-based checks are applied to detect suspicious patterns such as misleading keywords, lack of trust signals, or abnormal link structures.
- Domain-level signals such as HTTPS usage, domain type, and domain age are analyzed.
- Finally, external validation is performed by checking for negative indicators like scam-related mentions or weak online presence.

Based on all these signals, the system assigns a risk score and classifies the website into one of three categories: ALLOW, REVIEW, or BLOCK.

3. Website Analysis

Website 1: **https://manifestwaresoftware.com/web/**

Observations:

- The website has a structured layout with clear service-oriented content.
- Business-related sections such as services and company information are present.
- Contact-related information is available, which indicates legitimacy.
- The word “earn” appears, but not in a strongly suspicious or misleading context.
- The number of links is relatively high, which is common in content-heavy service websites.
- External validation signals are partially available, but WHOIS lookup was not successful.

Evaluation:

- Suspicious Keyword → Low severity
- High Link Count → Medium (contextually acceptable)
- External Signals → Partially available (WHOIS unavailable)

Final Result:

- Risk Score: 60–65
- Decision: REVIEW

Explanation:

Although the website shows multiple positive trust signals such as structured content and business-related information, some moderate-risk indicators were identified, including keyword presence and high link density. Additionally, the absence of WHOIS data introduces uncertainty in verifying domain credibility.

To avoid false approvals, the system takes a conservative approach and classifies this website as REVIEW, indicating that it may require manual verification before onboarding.

Website 2: https://zylotechindia.online/

Observations:

- The website has limited structured and professional content.
- The word "earn" appears in a more promotional and potentially misleading context.
- The domain uses ".online", which is often associated with lower credibility websites.
- Contact and business identity signals are weak or unclear.
- External presence is minimal, indicating low credibility.

Evaluation:

- Suspicious Keyword → Medium severity
- Domain Type (. online) → Medium risk
- Weak Trust Signals → High risk
- External Signals → Weak

Final Result:

- Risk Score: 75–85
- Decision: BLOCK

Explanation:

Multiple moderate-to-high risk indicators are present, including weak trust signals, domain credibility concerns, and limited external validation. These combined factors significantly increase the likelihood of the website being unreliable or potentially fraudulent.

Therefore, the system classifies this website as BLOCK, indicating high risk and recommending rejection during onboarding.

4. Conclusion

This system demonstrates a practical and explainable approach to website risk evaluation using rule-based logic and external validation signals.

It effectively distinguishes between relatively safe and potentially risky websites by combining internal content analysis with domain-level and reputation-based checks. The approach ensures transparency in decision-making while maintaining a balance between accuracy and simplicity.

5. Future Improvements

- Integrate real external APIs such as Google Search and WHOIS services for stronger validation
- Enhance keyword detection using context-aware techniques (NLP)
- Introduce domain reputation scoring and blacklist checks
- Improve scoring logic with dynamic weighting and confidence scoring